

****Harmonic Shape Transform (HST):**

Harmonic Note as an Intermediate Layer for Shape Transfer**

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Abstract

This document introduces a new concept for transforming shapes using the harmonic structure of a domain. Every shape can be associated with a harmonic note, defined as a normalized eigenfunction of the Laplace operator. These harmonic notes form smooth, proportional, and topologically stable fields that can serve as an intermediate representation between arbitrary shapes. I propose a new transformation method, the *Harmonic Shape Transform (HST)*, which maps shape A to shape B through their normalized harmonic notes. This approach enables smooth morphing between shapes with different topology, provides a new representation for machine learning, and offers potential applications in computer graphics, biomechanics, and shape theory.

1. Introduction

Shape transformation is a central problem in computer graphics, geometry processing, anatomy, and machine learning. Existing methods often rely on surface correspondence or mesh connectivity, which becomes problematic when shapes differ significantly in topology or geometry.

This document introduces a new approach:

Use a Laplace eigenfunction as a universal intermediate layer between shapes.

This eigenfunction — the *harmonic note* — is smooth, proportional, and independent of sharp boundaries or surface details. It can therefore serve as a stable “inner coordinate system” for transferring information between shapes.

2. Harmonic Note

Let $A \subset \mathbb{R}^n$ be a bounded domain. A harmonic note is defined as a normalized eigenfunction of the Laplace operator:

$$-\Delta u = \lambda u \text{ in } A,$$

with Dirichlet or Neumann boundary conditions.

The normalized note is:

$$\tilde{u} = \frac{u}{\|u\|_{L^2(A)}}.$$

Key properties:

- smooth inside the domain
- has natural level sets and nodal lines
- proportional (scale-invariant)
- captures the intrinsic harmonic structure of the shape

These properties make it suitable as an intermediate representation.

3. Definition of the Harmonic Shape Transform (HST)

Let A and B be two domains. Let $\tilde{u}$ be the harmonic note on A and $\tilde{v}$ the harmonic note on B .

Definition: The Harmonic Shape Transform $T_{A \rightarrow B}$ is a mapping:

$$T_{A \rightarrow B}: A \rightarrow B,$$

such that:

$$\tilde{v}(T_{A \rightarrow B}(x)) = \tilde{u}(x)$$

for all $x \in A$.

In other words:

Each point of shape A is mapped to the point in shape B that has the same harmonic note value.

This produces a smooth transformation independent of shape topology.

4. Applications

4.1 Computer Graphics

- Smooth morphing between shapes with different topology
- Texture transfer using harmonic level sets
- Animation without rigging
- Shape blending and interpolation

4.2 Machine Learning

- New embedding for shape representation
- Shape comparison using harmonic fields
- Generative models based on harmonic interpolation

4.3 Biomechanics and Anatomy

- Analysis of body proportions
- Comparison of anatomical structures
- Reconstruction of incomplete or deformed shapes
- Harmonic mapping of muscle chains

4.4 Mathematics

- New type of mapping between domains
- Spectral invariants
- Harmonic normalization of shapes

5. Originality of the Concept

Although ancient traditions (Kabbalah, Pythagorean philosophy, Hermeticism) work with symbolic ideas of internal harmony, none of them describe a mathematical principle of the form:

$$\text{shape} \rightarrow \text{harmonic note} \rightarrow \text{shape}.$$

This concept is new and does not appear in existing literature.

6. Conclusion

The Harmonic Shape Transform (HST) introduces a new method for transferring shapes through their internal harmonic structure. The approach is universal, smooth, and independent of topology. It has potential applications in computer graphics, machine learning, biomechanics, and theoretical mathematics.

7. Suggested Citation

If you wish to cite this concept:

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3. General Definition of the Harmonic Shape Transform (HST)

3.1. Scalar Field–Based Shape Transform

Let

$$\Omega_A, \Omega_B \subset \mathbb{R}^n$$

be two shapes (domains) equipped with smooth scalar fields

$$f_A: \Omega_A \rightarrow \mathbb{R}, f_B: \Omega_B \rightarrow \mathbb{R}.$$

These scalar fields serve as functional intermediaries between the shapes. They may represent, for example:

Laplace eigenfunctions,

heat kernel signatures (HKS),

wave kernel signatures (WKS),

signed distance functions (SDF),

curvature-based functions,

or any other smooth scalar descriptors.

To ensure comparability, we define normalized versions of these fields:

$$\tilde{f}_A = N_A(f_A), \tilde{f}_B = N_B(f_B),$$

where N_A, N_B are normalization operators (e.g., scaling to $[0,1]$, standardization, or monotonic reparameterization).

3.2. Definition of the General HST

The General Harmonic Shape Transform (HST) from Ω_A to Ω_B induced by scalar fields f_A, f_B is any mapping

$$T_{f_A \rightarrow f_B}: \Omega_A \rightarrow \Omega_B$$